

Design and Self-Assembly of Macrocycles with Metals at the Corners Based on Dissymmetric Terpyridine Ligands

Yaping Xu,^[a] Haoyue Su,^[a] Qixia Bai,^[b] Fang Fang,^[c] Jianjun Ma,^[a] Zhe Zhang,^[b] Xin-Qi Hao,^[d] Junjuan Shi,^{*[a]} Pingshan Wang,^[b] and Ming Wang^{*[a]}

Abstract: Terpyridine-based discrete supramolecular architectures with metal ions at the corners have rarely been reported. Herein, we report two dissymmetric terpyridyl ligands **LA** and **LB** decorated at the 5-position and 4-position of terpyridine respectively. The complexes constructed by the self-assembly of **LA** and **LB** with Zn(II) exhibit hand-circle-like

structures. Moreover, all Zn(II) are successfully fixed in the corners. A series of dimeric to hexameric macrocycles is obtained by head-to-tail connections with changing concentration. This work will pave the way for preparation of more elaborate self-assembled structures based on dissymmetric ligands.

Introduction

Nature has created many complex and functional biological systems with unique structures and desirable functions by assembling simple building blocks. Inspired by the powerful bottom-up self-assembly approach in Nature, chemists have endeavored to develop novel supramolecular architectures by exploiting various non-covalent interactions,^[1] such as hydrogen bonding, metal-coordination interactions, dipole-dipole interaction, etc. Among them, coordination-driven self-assembly has received considerable attention^[2] and been proven to be a well-established methodology for constructing discrete supramolecular structures, due to the high predictability and controllability of metal-ligand interactions.^[3] Over the past few decades, a variety of two-dimensional (2D)^[4] and three-dimensional (3D)^[5] metallo-supramolecules with precisely controlled shape, size, and stoichiometry has been successfully obtained by different types of ligand-metal interactions, including the coordination from pyridine,^[6] bipyridine,^[7] and terpyridine

(tpy)^[8] with end-capped metal components or naked metal ions. Moreover, the used ligands and metal ions could have an accurate distribution as expected, such as the metal ions can be placed accurately at the edges or corners of obtained structures.

2,2':6',2''-tpy and its derivatives, possessing strong coordination ability with a variety of naked transition metal ions,^[9] have been widely employed to construct multiple kinds of metal-organic cycles^[10] and metal-organic cages^[11] with unique properties and wide applications, e.g., luminescence,^[12] self-healing,^[13] catalysis,^[14] and antibacterial.^[2e,15] However, in all reported supramolecular architectures constructed with tpy units, metal ions were mainly designed in the edge rather than corner of the final structures,^[5a] due to limitations of linear < tpy-metal-tpy > connectivity and the position of the tpy in the ligand. Therefore, the organic directing units used for linking tpy play a decisive role in configuration of the final structure. To construct tpy-based discrete supramolecular architectures with metal ions as corners, the question needed to be solved is how can we reasonably design the ligands.

Herein, we reported the design, synthesis, and self-assembly of two dissymmetric ditopic tpy ligands (**LA** and **LB**) with Zn(II) (Scheme 1). The unique design of the two ditopic ligands is that the one tpy unit was attached to the 5-position or 4-position of another tpy unit. Such molecular design makes two types of tpy within one ligand have significant coordination selectivity. By head-to-tail connectivity, dimeric (**Zn₂LA₂**) and tetrameric (**Zn₄LB₄**) macrocycles with metal ions as corners can be obtained by self-assembly of the dissymmetric ligands **LA** and **LB** with Zn(II) respectively at a relatively low concentration. Although larger size macrocycles, such as trimer **Zn₃LA₃**, pentamer **Zn₅LB₅**, and hexamer **Zn₆LB₆**, were found in the self-assembly system of **LA** or **LB** with increasing the concentration, metal ions Zn(II) were still fixed at the corners of macrocycles.

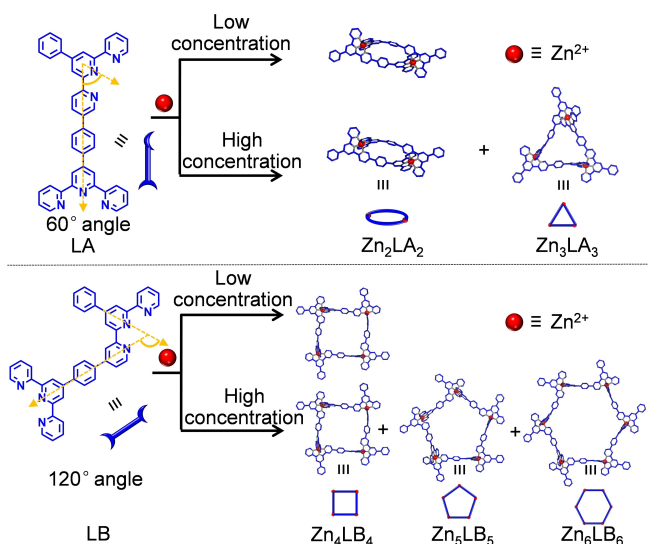
[a] Y. Xu, H. Su, J. Ma, Dr. J. Shi, Prof. Dr. M. Wang
State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry
Jilin University
Changchun, Jilin 130012, P. R. China
E-mail: mingwang358@jlu.edu.cn
jjshi18@mails.jlu.edu.cn

[b] Q. Bai, Dr. Z. Zhang, Prof. Dr. P. Wang
Institute of Environmental Research at Greater Bay Area, Key Laboratory for Water Quality and Conservation of the Pearl River Delta, Ministry of Education
Guangzhou University
Guangzhou, Guangdong 510006, P. R. China

[c] Dr. F. Fang
Instrumental Analysis Center
Shenzhen University
Shenzhen, Guangdong 518055, P. R. China

[d] Prof. Dr. X.-Q. Hao
College of Chemistry and Green Catalysis Center
Zhengzhou University
Zhengzhou Henan 450001, P. R. China

Supporting information for this article is available on the WWW under <https://doi.org/10.1002/asia.202200071>



Scheme 1. Self-assembly of macrocycles with dissymmetric ligands and metals at the corners.

Results and Discussion

The dissymmetric ditopic terpyridyl ligand **LA** was designed and synthesized by a Suzuki Coupling reaction of the dissymmetric precursor (compound **4**) and symmetric compound in a good yield (80%). The symmetric compound [4'-(4-Boronophenyl)-2,2':6',2''-terpyridine] was synthesized through one step condensation according to the reported literature.^[16] It is a critical step that the obtaining of key dissymmetric compound **4**, bearing one bromine at the 5-position of tpy, which was synthesized by a typical Kröhnke reaction^[17] (Scheme S1). **LA** and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a stoichiometric ratio (1:1) were mixed in CHCl_3 and MeOH (1:3, v/v) at 50°C for 10 h. After that, excessive NH_4PF_6 (to exchange NO_3^- with PF_6^-) in methanol was added to give a white precipitate (**SA**).

Ligand **LA** and the precipitate **SA** were first characterized by NMR spectroscopy. Given the dissymmetry of ligand **LA**, five kinds of pyridine signals were observed by two-dimensional correlation spectroscopy (2D-COSY). ^1H NMR of precipitate **SA** at low concentration (0.5 mg/mL) also showed the expected five kinds of pyridines signals, suggesting the formation of a highly symmetric structure. As shown in Figure 1, after coordination with $\text{Zn}(\text{II})$, three significant upfield shifts for protons of A^6 , $\text{A}^{6''}$, and $\text{B}^{6,6''}$ were clearly observed with $\Delta\delta = 2.6$, 0.4, and 0.9 ppm, respectively. The protons in A^3 , $\text{B}^{3,3''}$, Ph^d , and Ph^e also showed slightly upfield shifts owing to the shielding effect. While other protons showed downfield shifts. These results supported the quantitative formation of a single-component complex. When increasing the concentration (2 mg/mL) of **SA**, a new set of peaks was observed (Figure 1c), indicating another complex was formed.

Electrospray ionization mass spectra (ESI-MS) as an effective method were further used to determine the compositions of **SA**. In ESI-MS spectrum (Figure 2a) of **SA** (0.5 mg/mL), one prominent set of peaks with charge states from 2+ to 4+ was

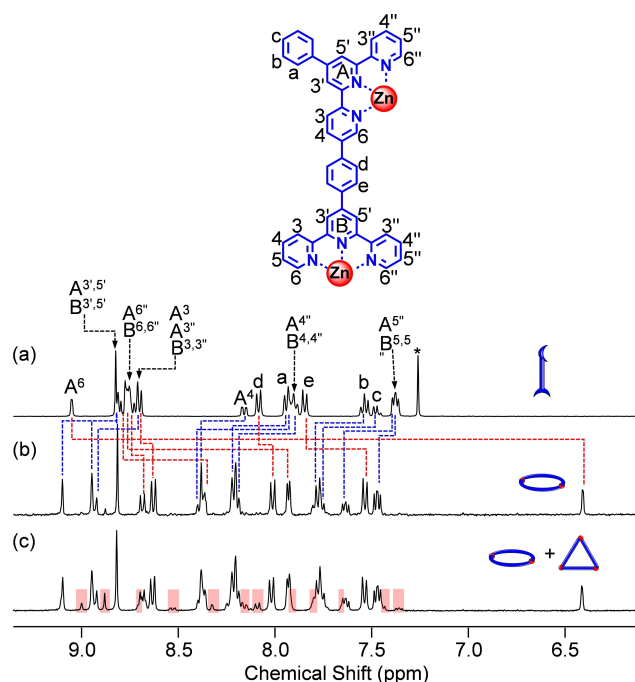


Figure 1. (a) ^1H NMR spectrum (500 MHz, CDCl_3 , 300 K) of ligand **LA**. ^1H NMR spectra (500 MHz, CD_3CN , 300 K) of (b) complex Zn_2LA_2 and (c) mixture of complexes Zn_2LA_2 and Zn_3LA_3 at a concentration of 0.5 mg/mL and 2 mg/mL respectively.

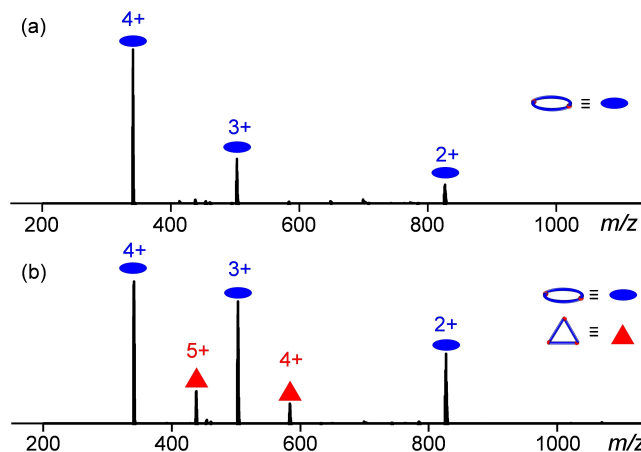


Figure 2. ESI-MS spectra of (a) complex Zn_2LA_2 (0.5 mg/mL) and (b) mixture of complexes Zn_2LA_2 and Zn_3LA_3 (2 mg/mL).

observed, due to the loss of a different number of PF_6^- . All the experimental isotope patterns closely matched the corresponding simulated isotope patterns (Figure S15). After deconvolution, the molecular weight of **SA** was 1944.06 Da, matching well with chemical composition of two **LA**, two $\text{Zn}(\text{II})$ ions and four PF_6^- . The results showed that the obtained supramolecule was the dimeric macrocycle Zn_2LA_2 , rather than trimer Zn_3LA_3 , indicating that organic ligands are far more flexible than we expected. When increasing the concentration (2 mg/mL) of **SA**, the other set of peaks was observed. After deconvolution based on m/z , its molecular weight (2916.09 Da) was exactly same as

that of the trimeric structure, indicating the composition of **SA** is concentration-dependent.

After successful characterization of the structural composition of **SA** based on dissymmetric ditopic ligand **LA**, we further adjusted the angle of two tpy units and obtained ligand **LB** decorated at the 4-position of tpy. Compared with **LA** (60° angle), the angle of two tpy units in **LB** was designed as 120°, which was an ideal angle for building a hexagon. Similar with **LA**, the key intermediate compound **7** was synthesized via a typical Kröhnke reaction. The dissymmetric ditopic terpyridyl ligand **LB** was synthesized by the same synthetic method as **LA** in a good yield (83%). The structure of ligand **LB** was established by NMR spectroscopy (¹H NMR, ¹³C NMR, and 2D COSY spectra) and HR ESI-MS analysis (Figure S2). Following the

same procedure as **LA**, **LB** coordinated with Zn(II) to give white solid **SB**.

NMR was first utilized to address the structural information about **SB**. When the acetonitrile solution of **SB** was diluted to the concentration of 0.5 mg/mL, ¹H NMR showed sharp and clear peaks as expected (Figure 3b). The complete assignments were further confirmed by 2D COSY experiments (Figure S12). Similar with complex **Zn₂LA₂**, **SB** (0.5 mg/mL) also showed five kinds sets of pyridines signals. The protons of A⁶, A^{6'}, and B^{6,6'} all showed upfield shifts with 0.6 ppm, 0.6 ppm, and 0.9 ppm, respectively, which is the typical feature of octahedral complex < tpy-M-tpy > [M=Zn(II)]. Protons in A⁵ also showed slightly upfield with 0.08 ppm. Protons in 3',5', 4,4", and 5,5" of all tpy showed obvious downfield shifts, confirming the formation of a discrete supramolecule. With increasing the concentration of **SB** to 2 mg/mL, the ¹H NMR showed broad and complicated peaks, meaning that a mixture or large structure was formed.

ESI-MS was then employed to identify the structural information on **SB**. As shown in Figure 4a, pure tetramers **Zn₄LB₄** was observed at a concentration of 0.5 mg/mL, showing a set of peaks with different charge states (4+ to 6+) for losing different number of PF₆⁻. When the concentration of **SB** was increased to 2 mg/mL, a mixture of tetramers, pentamers and hexamers was obtained, named as **Zn₄LB₄**, **Zn₅LB₅**, and **Zn₆LB₆** respectively. From the ESI-MS results, the obtained molecular weight of **Zn₄LB₄**, **Zn₅LB₅**, and **Zn₆LB₆** was 3888.12, 4860.15, and 5832.18 Da, respectively. The experimental isotope patterns of these three assemblies all agreed well with the corresponding simulated isotope patterns (Figures S17–S20). Moreover, the system of a mixture with three kinds of macrocycles suggested that the complex constructed by **LA** had higher thermodynamic stability than **LB**.

According to the all of the above information, energy-minimized structures (Figure 5) from molecular modeling of these macrocycles were obtained. Due to the dissymmetry of

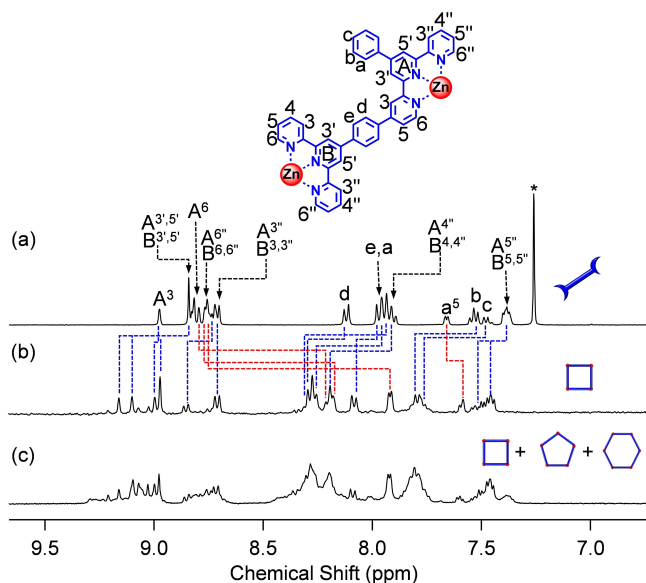


Figure 3. ¹H NMR spectrum (400 MHz, CDCl₃, 300 K) of (a) ligand **LA**. ¹H NMR spectra (400 MHz, CD₃CN, 300 K) of (b) complex **Zn₄LB₄** and (c) mixture of complexes **Zn₄LB₄**, **Zn₅LB₅**, and **Zn₆LB₆** at a concentration of 0.5 mg/mL and 2 mg/mL respectively.

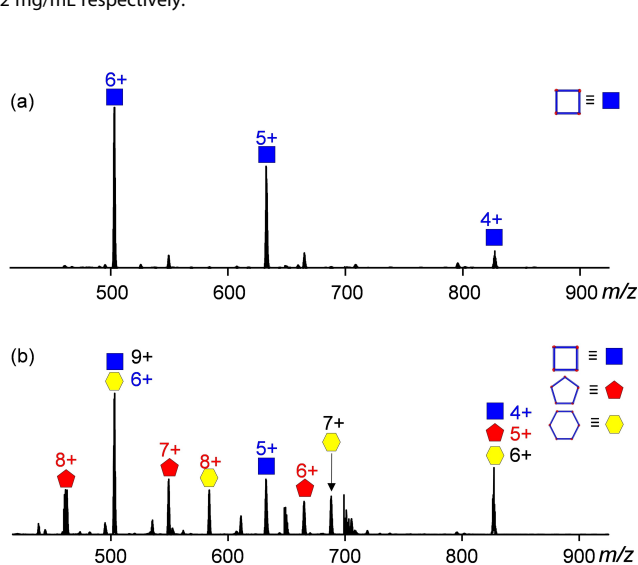


Figure 4. ESI-MS spectra of (a) complex **Zn₄LB₄** (0.5 mg/mL) and (b) mixture of **Zn₄LB₄**, **Zn₅LB₅**, and **Zn₆LB₆** (2 mg/mL).

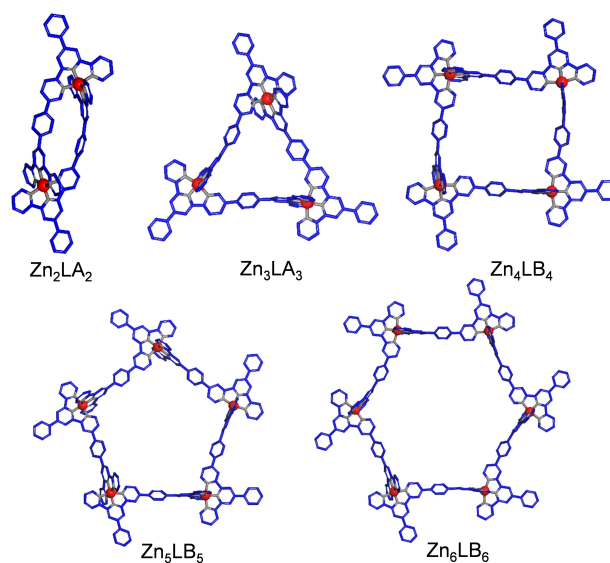


Figure 5. Energy-minimized structures from molecular modeling of **Zn₂LA₂**, **Zn₃LA₃**, **Zn₄LB₄**, **Zn₅LB₅**, and **Zn₆LB₆**.

ligands, LA and LB showed coordination selectivity and coordinated with Zn(II) in a style of head-to-tail coordination mode to construct a series of macrocycles. Therefore, all metal ions are placed at the corners of the macrocycles.

Conclusion

In summary, we designed and synthesized two dissymmetric ditopic tpy ligands LA and LB with different angles, which are caused by substitution of the tpy at the 5-position or 4-position. In the process of self-assembly, the dissymmetric structures showed high selectivity and head-to-tail coordination mode. Different from the previous reported structures based on tpy units, the metal ions can be easily fixed at the corners of macrocycles. Moreover, these structures showed concentration dependence. This strategy of designing dissymmetric ligand will pave a new avenue for the development of metallo-supramolecular architectures with unique geometry and high selectivity.

Experimental Section

General Procedures. All reagents were purchased from Sigma-Aldrich and Alfa Aesar and were used without further purification. Column chromatography was conducted using SiO₂ (VWR, 40–60 μm, 60 Å) and the separated products were visualized by UV light. All solvents for palladium-catalyzed cross-coupling reactions were degassed with nitrogen.

Nuclear magnetic resonance (NMR). NMR spectra data were recorded on Bruker AVANCEIII 500 MHz and 400 MHz nuclear magnetic resonance instruments using CDCl₃, DMSO-*d*₆, or CD₃CN as the solvent. Chemical shifts were given in ppm using residue solvent signal as the reference.

Electrospray ionization-mass spectrometry (ESI-MS). Electrospray ionization (ESI) mass spectra of complexes were recorded with a Waters Synapt G2 tandem mass spectrometer, using solutions of 0.5 mg or 2 mg sample in 1 mL of MeCN/MeOH (3:1, v/v) for complexes. HR ESI-MS of ligands were recorded on a Bruker microTOF-Q II Mass Spectrometer.

Molecular Modeling. Energy-minimized structures were obtained with Materials Studio 8.0 program (Accelrys Software, Inc.) using universal force field. The parameters were set according to literature.^[18]

Acknowledgements

We gratefully acknowledge the support from the National Natural Science Foundation of China (22071079 for M.W.).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Self-assembly • Metallo-supramolecule • Terpyridine • Macrocycle

- [1] a) L.-J. Chen, H.-B. Yang, *Acc. Chem. Res.* **2018**, *51*, 2699–2710; b) D. J. Cram, *Angew. Chem. Int. Ed.* **1988**, *27*, 1009–1020; c) P. A. Gale, E. N. W. Howe, X. Wu, *Chem* **2016**, *1*, 351–422; d) J. M. Lehn, *Pure Appl. Chem.* **1978**, *50*, 871–892; e) H. Li, H. C. Zhang, A. D. Lammer, M. Wang, X. P. Li, V. M. Lynch, J. L. Sessler, *Nat. Chem.* **2015**, *7*, 1003–1008; f) C. J. Pedersen, *J. Am. Chem. Soc.* **1967**, *89*, 7017–7036; g) R. W. Saalfrank, H. Maid, A. Scheurer, *Angew. Chem. Int. Ed.* **2008**, *47*, 8794–8824; h) Y. Sun, C. Y. Chen, P. J. Stang, *Acc. Chem. Res.* **2019**, *52*, 802–817; i) S. Zarra, D. M. Wood, D. A. Roberts, J. R. Nitschke, *Chem. Soc. Rev.* **2015**, *44*, 419–432.
- [2] a) M. Barboiu, A.-M. Stadler, J.-M. Lehn, *Angew. Chem. Int. Ed.* **2016**, *55*, 4130–4154; b) A. K. W. Chan, V. W. W. Yam, *Acc. Chem. Res.* **2018**, *51*, 3041–3051; c) Y. S. Chen, E. Solel, Y.-F. Huang, C.-L. Wang, T.-H. Tu, E. Keinan, Y.-T. Chan, *Nat. Commun.* **2019**, *10*; d) T. R. Cook, P. J. Stang, *Chem. Rev.* **2015**, *115*, 7001–7045; e) I. Eryazici, C. N. Moorefield, G. R. Newkome, *Chem. Rev.* **2008**, *108*, 1834–1895; f) R. S. Forgan, J.-P. Sauvage, J. F. Stoddart, *Chem. Rev.* **2011**, *111*, 5434–5464; g) E. G. Percastegui, T. K. Ronson, J. R. Nitschke, *Chem. Rev.* **2020**, *120*, 13480–13544; h) M. Ruben, J. Rojo, F. J. Romero-Salguero, L. H. Uppadine, J. M. Lehn, *Angew. Chem. Int. Ed.* **2004**, *43*, 3644–3662; i) J. Shi, M. Wang, *Chem. Asian J.* **2021**, *16*, 4037–4048; j) H. Wang, Y. Li, N. Li, A. Filosa, X. Li, *Nat. Rev. Mater.* **2021**, *6*, 145–167; k) T. Wu, Z. Jiang, Q. Bai, Y. Li, S. Mao, H. Yu, L. Wojtas, Z. Tang, M. Chen, Z. Zhang, T.-Z. Xie, M. Wang, X. Li, P. Wang, *Chem* **2021**, *7*, 2429–2441; l) M. Yoshizawa, J. K. Klosterman, M. Fujita, *Angew. Chem. Int. Ed.* **2009**, *48*, 3418–3438.
- [3] a) R. Chakraborty, P. S. Mukherjee, P. J. Stang, *Chem. Rev.* **2011**, *111*, 6810–6918; b) D. Fujita, Y. Ueda, S. Sato, N. Mizuno, T. Kumasaka, M. Fujita, *Nature* **2016**, *540*, 563–566; c) J. M. Lehn, *Chem. Soc. Rev.* **2007**, *36*, 151–160; d) B. Olenyuk, J. A. Whiteford, A. Fechtenkotter, P. J. Stang, *Nature* **1999**, *398*, 796–799.
- [4] a) J.-F. Ayme, J. E. Beves, C. J. Campbell, D. A. Leigh, *Chem. Soc. Rev.* **2013**, *42*, 1700–1712; b) G. Gil-Ramirez, D. A. Leigh, A. J. Stephens, *Angew. Chem. Int. Ed.* **2015**, *54*, 6110–6150; c) S. Leininger, B. Olenyuk, P. J. Stang, *Chem. Rev.* **2000**, *100*, 853–907; d) Y. Li, H. Wang, X. Li, *Chem. Sci.* **2020**, *11*, 12249–12268; e) S.-Y. Wang, J.-Y. Huang, Y.-P. Liang, Y.-J. He, Y.-S. Chen, Y.-Y. Zhan, S. Hiraoka, Y.-H. Liu, S.-M. Peng, Y.-T. Chan, *Chem. Eur. J.* **2018**, *24*, 9274–9284.
- [5] a) S. Chakraborty, G. R. Newkome, *Chem. Soc. Rev.* **2018**, *47*, 3991–4016; b) T. Sawada, M. Fujita, *Chem* **2020**, *6*, 1861–1876; c) M. M. J. Smulders, I. A. Riddell, C. Browne, J. R. Nitschke, *Chem. Soc. Rev.* **2013**, *42*, 1728–1754.
- [6] a) D. K. Chand, K. Biradha, M. Kawano, S. Sakamoto, K. Yamaguchi, M. Fujita, *Chem. Asian J.* **2006**, *1*, 82–90; b) C. Mu, Z. Zhang, Y. Hou, H. Liu, L. Ma, X. Li, S. Ling, G. He, M. Zhang, *Angew. Chem. Int. Ed.* **2021**, *60*, 12293–12297; c) J. B. Pollock, T. R. Cook, G. L. Schneider, P. J. Stang, *Chem. Asian J.* **2013**, *8*, 2423–2429; d) Q. F. Sun, J. Iwasa, D. Ogawa, Y. Ishido, S. Sato, T. Ozeki, Y. Sei, K. Yamaguchi, M. Fujita, *Science* **2010**, *328*, 1144–1147; e) B. Wu, Z. Guo, G. Li, J. Zhao, Y. Liu, J. Wang, H. Wang, X. Yan, *Chem. Commun.* **2021**, *57*, 11056–11059.
- [7] a) J. J. Danon, A. Kruger, D. A. Leigh, J. F. Lemonnier, A. J. Stephens, I. J. Vitorica-Yrezabal, S. L. Woltering, *Science* **2017**, *355*, 159–162; b) H.-J. Nie, C.-J. Yao, J. Yao, Y.-W. Zhong, *Chem. Asian J.* **2011**, *6*, 3322–3327.
- [8] G. R. Newkome, P. S. Wang, C. N. Moorefield, T. J. Cho, P. P. Mohapatra, S. N. Li, S. H. Hwang, O. Lukoyanova, L. Echegoyen, J. A. Palagallo, V. Iancu, S. W. Hla, *Science* **2006**, *312*, 1782–1785.
- [9] a) E. C. Constable, *Chem. Soc. Rev.* **2007**, *36*, 246–253; b) S. De, K. Mahata, M. Schmittel, *Chem. Soc. Rev.* **2010**, *39*, 1555–1575; c) H. Hofmeier, U. S. Schubert, *Chem. Soc. Rev.* **2004**, *33*, 373–399; d) U. S. Schubert, C. Eschbaumer, *Angew. Chem. Int. Ed.* **2002**, *41*, 2892–2926; e) A. Wild, A. Winter, F. Schlutter, U. S. Schubert, *Chem. Soc. Rev.* **2011**, *40*, 1459–1511.
- [10] a) J. Ma, T. Lu, X. Duan, Y. Xu, Z. Li, K. Li, J. Shi, Q. Bai, Z. Zhang, X.-Q. Hao, Z. Chen, P. Wang, M. Wang, *Commun. Chem.* **2021**, *4*, 1–11; b) B.

- Song, S. Kandapal, J. Gu, K. Zhang, A. Reese, Y. Ying, L. Wang, H. Wang, Y. Li, M. Wang, S. Lu, X.-Q. Hao, X. Li, B. Xu, X. Li, *Nat. Commun.* **2018**, 9, 4575; c) Y.-C. Wang, Y.-P. Liang, J.-Y. Cai, Y.-J. He, Y.-H. Lee, Y.-T. Chan, *Chem. Commun.* **2016**, 52, 12622–12625.
- [11] a) M. Z. Chen, J. Wang, D. Liu, Z. L. Jiang, Q. Q. Liu, T. Wu, H. S. Liu, W. D. Yu, J. Yan, P. S. Wang, *J. Am. Chem. Soc.* **2018**, 140, 2555–2561; b) X. Jiang, H. Yu, J. Shi, Q. Bai, Y. Xu, Z. Zhang, X.-Q. Hao, B. Li, P. Wang, L. Wu, M. Wang, *CCS Chem.* **2021**, 2042–2054.
- [12] G.-Q. Yin, H. Wang, X.-Q. Wang, B. Song, L.-J. Chen, L. Wang, X.-Q. Hao, H.-B. Yang, X. P. Li, *Nat. Commun.* **2018**, 9, 567.
- [13] Y. Shi, M. Wang, C. Ma, Y. Wang, X. Li, G. Yu, *Nano Lett.* **2015**, 15, 6276–6281.
- [14] C. Y. Wei, Y. He, X. D. Shi, Z. G. Song, *Coord. Chem. Rev.* **2019**, 385, 1–19.
- [15] H. Wang, X. Qian, K. Wang, M. Su, W. W. Haoyang, X. Jiang, R. Brzozowski, M. Wang, X. Gao, Y. Li, B. Xu, P. Eswara, X.-Q. Hao, W. Gong, J. L. Hou, J. Cai, X. Li, *Nat. Commun.* **2018**, 9, 1815.
- [16] Z. Zhang, H. Wang, X. Wang, Y. Li, B. Song, O. Bolarinwa, R. A. Reese, T. Zhang, X. Q. Wang, J. Cai, B. Xu, M. Wang, C. Liu, H.-B. Yang, X. Li, *J. Am. Chem. Soc.* **2017**, 139, 8174–8185.
- [17] F. Kröhnke, *Synthesis* **1976**, 1–24.
- [18] M. Wang, C. Wang, X.-Q. Hao, J. Liu, X. Li, C. Xu, A. Lopez, L. Sun, M.-P. Song, H.-B. Yang, X. Li, *J. Am. Chem. Soc.* **2014**, 136, 6664–6671.

Manuscript received: January 25, 2022
 Revised manuscript received: February 23, 2022
 Accepted manuscript online: February 24, 2022
 Version of record online: March 4, 2022
